

WEST FREUGH, United Kingdom

WMO: 031320

Lat: 54.8592N

Lon: 4.9353W

Elev: 39

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.5	28.3	22.5	16.9	30.5	24.7	18.8	32.1	39.8	44.7	34.8	45.0	2.6	320	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	70.8	62.5	67.5	60.8	65.2	59.6	63.9	68.6	62.2	65.8	60.9	63.7	10.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	83.1	65.8	60.7	79.3	63.7	59.5	76.0	62.5	29.2	68.9	28.0	66.1	27.0	63.8	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.0	27.6	24.8	21.5	76.4	2.9	3.7	19.4	79.1	17.7	81.2	16.1	83.3	14.0	86.0
			21.2	66.9	3.6	2.1	18.6	68.4	16.5	69.6	14.5	70.8	11.9	72.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	41.0	41.1	42.7	46.0	50.5	55.0	58.0	57.8	54.9	50.4	45.5	41.6
	DBStd	7.75	5.07	4.74	4.52	4.30	4.64	3.61	3.53	3.34	3.85	4.75	5.21	6.01
	HDD50	1414	281	250	228	132	49	3	0	0	6	55	149	261
	HDD65	5933	745	669	691	571	450	299	220	224	302	452	586	724
	CDD50	958	0	1	1	12	64	154	248	242	153	68	13	2
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	30	0	0	0	0	3	5	19	3	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	11.5	13.7	13.5	12.4	11.4	10.6	10.1	9.3	9.8	10.4	12.0	12.4	13.0
Precipitation	PrecAvg	38.5	4.0	2.5	2.8	2.4	2.2	2.4	2.6	3.5	3.9	4.2	4.2	3.8
	PrecMax	47.8	6.5	6.0	6.4	5.6	5.2	4.3	6.0	6.5	6.5	7.5	7.2	6.9
	PrecMin	30.0	0.6	0.0	1.1	0.4	0.6	0.9	0.7	0.9	0.6	0.9	2.0	0.6
	PrecStd	4.4	1.5	1.3	1.3	1.1	1.2	0.8	1.3	1.3	1.7	1.6	1.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	52.8	55.8	63.9	72.3	73.6	76.8	72.7	67.3	61.7	57.4	54.6
		MCWB	50.6	50.3	50.3	53.9	60.1	62.0	66.0	64.8	61.5	58.0	55.6	52.9
	2%	DB	51.2	51.0	53.1	58.2	66.0	68.0	71.3	68.2	64.5	59.8	55.7	53.1
		MCWB	49.4	48.9	49.1	50.9	57.2	60.3	63.5	62.0	59.6	57.0	53.8	51.7
	5%	DB	49.9	49.5	51.1	55.1	61.8	64.8	67.6	65.7	62.8	58.7	54.4	51.8
		MCWB	48.4	47.4	47.8	49.7	55.0	58.7	61.6	60.4	58.6	56.1	52.6	50.3
	10%	DB	48.5	48.2	49.6	53.0	58.6	62.4	64.9	63.9	61.4	57.5	53.2	50.2
		MCWB	46.8	46.2	46.8	48.6	53.1	57.1	59.9	59.2	57.7	55.0	51.3	48.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	51.4	51.9	54.6	61.6	64.0	67.0	65.7	63.3	59.2	56.3	53.6
		MCDB	52.2	52.2	54.1	61.4	69.1	70.5	74.4	70.7	65.8	60.4	57.0	54.1
	2%	WB	49.9	49.4	50.0	52.3	58.5	61.5	64.4	63.4	61.2	58.0	54.3	52.0
		MCDB	50.9	50.5	52.1	56.5	64.7	66.8	69.8	67.0	63.0	59.3	55.2	52.9
	5%	WB	48.6	48.1	48.7	50.8	56.0	59.6	62.5	61.8	59.8	56.8	52.9	50.5
		MCDB	49.7	49.3	50.4	53.8	60.5	63.9	66.7	64.3	61.7	58.3	54.1	51.6
	10%	WB	47.1	46.6	47.4	49.4	53.9	57.9	60.8	60.6	58.5	55.6	51.6	48.8
		MCDB	48.4	48.1	49.2	52.1	57.5	61.4	63.8	62.6	60.5	57.2	53.1	50.1
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.4	11.3	12.7	14.1	13.2	13.0	12.7	12.2	11.0	9.5	9.4
		MCDBR	8.2	8.6	11.9	15.4	19.1	17.3	17.8	14.9	13.6	10.6	8.7	8.9
		MCWBR	8.3	7.8	9.0	10.4	12.4	11.2	11.4	9.9	9.7	8.9	8.7	9.1
	5% WB	MCDBR	8.3	7.9	10.6	13.2	16.5	16.1	15.4	12.0	11.3	9.0	8.4	9.0
		MCWBR	8.5	7.8	8.9	9.5	11.4	11.1	10.6	8.8	9.0	8.3	8.7	9.3
Clear-Sky Solar Irradiance	taub	0.334	0.345	0.369	0.386	0.390	0.384	0.384	0.370	0.363	0.348	0.327	0.320	
	taud	2.256	2.341	2.319	2.309	2.350	2.399	2.399	2.400	2.453	2.469	2.440	2.331	2.228
	Ebn at Noon	184	232	253	266	271	274	271	269	255	229	188	160	
	Edh at Noon	19	24	31	36	36	35	34	31	27	22	17	16	
All-Sky Solar Radiation	RadAvg	170	386	760	1229	1573	1574	1505	1237	858	468	222	126	
	RadStd	20	40	58	91	136	157	133	91	58	54	23	13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page